

Lecture Notes no 11 for a Three Factorial Completely Randomized Fixed Effects Design

This lecture presents a statistical data analysis for the data set regarding the soft drink bottling experiment described in Example 5.3, in Section 5.4, pages 174-179, in our textbook. The design in this experiment is a completely randomized three factorial fixed effects design. The three factors in this experiment are: Percent Carbonation (A) with three levels, Operating Pressure (B) with two levels and Line Speed (C) with two levels. Two replicates have been obtained for each of the 12 treatment combinations. The response variable is the average deviation from the target fill height observed in the production run at each set of conditions. The model for this experiment is:

$$Y_{ijkl} = \mu + A_i + B_j + C_k + (AB)_{ij} + (AC)_{ik} + (BC)_{jk} + (ABC)_{ijk} + \varepsilon_{l(ijk)},$$

where $i = 1, 2, 3$, $j = 1, 2$, $k = 1, 2$ and $l = 1, 2$. In this experiment we are interested in testing the following seven null hypotheses:

$$H_{01} : A_i = 0, i = 1, 2, 3,$$

$$H_{02} : B_j = 0, j = 1, 2,$$

$$H_{03} : C_k = 0, k = 1, 2,$$

$$H_{04} : AB_{ij} = 0, i = 1, 2, 3, j = 1, 2,$$

$$H_{05} : AC_{ik} = 0, i = 1, 2, 3, k = 1, 2,$$

$$H_{06} : BC_{jk} = 0, j = 1, 2, k = 1, 2,$$

$$H_{07} : ABC_{ijk} = 0, i = 1, 2, 3, j = 1, 2, k = 1, 2.$$

If one decides on the overall significance level to be .15, then the nominal significance level to be used for each test is going to be $.15/7 \approx .02143$. This means that we will reject a null hypothesis only if the associated F statistic will have a p-value that is less than .02143.

The SAS program file for the model given above is SAS file no 1 for soft drink Example 5.3 and its output is SAS output no 1 for soft drink Example 5.3. Note that on page 3 in this SAS output, the normal probability plot has streaks of straight lines parallel to the horizontal axis. Moreover, the W statistic, on page 14 of the SAS output, has a p-value of $.0039 < .01$, which indicates that the normality assumption of the random errors is not valid.

Note that the statement made in the textbook on page 177, second paragraph after the equation for S_E , is **not correct**: " point out that a normal probability plot of the residuals and other diagnostics do not indicate any major problem." As we have seen above, the normality assumption for the random errors is not valid!

Therefore, we cannot use this SAS output to analyze this experiment. One approach to overcome this problem, is to transform the data. The following transformation has been used in SAS file no 2 for soft drink Example 5.3:

$$Y_{ijkl}^* = \log(Y_{ijkl} + 20).$$

The reason we have added a positive constant to the observed data, before applying the log transformation, is that the original data has negative numbers. One could try other positive constants that will assure that the underlying assumptions, for the model with transformed data, are satisfied. Therefore, we now can proceed to analyze the following model:

$$Y_{ijkl}^* = \mu + A_i + B_j + C_k + (AB)_{ij} + (AC)_{ik} + (BC)_{jk} + (ABC)_{ijk} + \varepsilon_{l(ijk)}.$$

First we examine the underlying assumptions for this model, based on SAS output no 2 for soft drink Example 5.3. It follows from the model diagnostics displayed on page 3 of this SAS output, that all the underlying assumptions for our model with transformed data are valid. There are no patterns in the plot of the residuals vs. predicted values (even though the plot looks somewhat structured). The normal probability plot looks fine, there are no unusually large or small RSTUDENT values and all Cook's D values are less than 1. The p-value of the W statistic is .1166 (page 14 of the output). Hence, we can proceed to examine the Type III SS table of F statistics and their p-values, given on page 2 of the SAS output no 2 for soft drink Example 5.3.

Based on the table of the F statistics and their p-values, we can reach the following conclusions: $H_{01} : A_i = 0, i = 1, 2, 3$ can be rejected since the F statistic has a value of 157.99 and its associated p-value is less than .0001, $H_{02} : B_j = 0, j = 1, 2$, can also be rejected since the F statistic has a value of 54.98 and its associated p-value is less than .0001, the $H_{03} : C_k = 0, k = 1, 2$, is also rejected as the F statistic has a value of 27.82 and its p-value is less than .0002. All other 4 null hypotheses are accepted as the p-values of their F statistics are larger than .02143. In fact, they all are larger than .4.

Since the interactions are not significant, we can proceed with analyzing the Tukey multiple comparisons displays. From the Tukey multiple comparison display for Percent Carbonation, we observe that the 3 means of the deviation overfill are significantly different from each other. When we convert to the original units (by exponentiating the values of the means and then subtracting 20) we find that for 14% carbonation the overfill deviation mean is 7.28, for 12% it is 2.42 and for the 10% it is -.54. Since -.54 is closest to 0, the 10% carbonation is the optimal level. Similarly, when converting to the original units the mean deviation overfill for the two levels of Operating Pressure, we find that for the 30 level we get a mean of 4.19 deviation overfill and for the 25 level we get an mean of 1.55 deviation overfill. Therefore, one would choose the 25 level of operating pressure. As far as line speed is concerned, for the 250 level one gets a mean overfill deviation of 3.79 (in the original units) and for the 200 level one gets a mean deviation overfill of 1.91 (in the original units). Therefore the 200 level of line speed is preferred.